ಬಿ.ಎಂ.ಎಸ್. ತಾಂತ್ರಿಕ ಮತ್ತು ವ್ಯವಸ್ಥಾಪನಾ ಮಹಾವಿದ್ಯಾಲಯ
BMS INSTITUTE OF TECHNOLOGY & MANAGEMENT
(Autonomous Institution Under VTU)
Yelahanka, Bengaluru -560119

DEPARTMENT OF CIVIL ENGINEERING

COURSE PROJECT: RAINFALL PREDICTION USING MACHINE LEARNING

COURSE: WATER CONSERVATION AND RAINWATER HARVESTING (BCV605A)

COURSE COORDINATORS: PROF ARCHANA K & PROF SHIMNA M

District of study: ______________________________

Student Name	Student USN	Course Coordinator	Marks

Submitted on:

Signature of course coordinator:

1. Instructions:

In this course project, students will model and predict future rainfall based on historical data using machine learning techniques on the Python platform. The goal is to develop accurate predictive models, understand trends in rainfall data, and improve skills in machine learning and time-series analysis.

Task Overview:

1. **Data Pre-processing:** Clean and prepare the given dataset for analysis.
2. **Feature Engineering:** Extract useful features such as time-related variables (e.g., year, month, etc.) and possibly other weather factors.
3. **Model Selection and Training:** Choose and train appropriate machine learning models (e.g., decision trees, random forests, neural networks, etc.).
4. **Model Evaluation:** Assess the performance using evaluation metrics like RMSE, R^2 score, and accuracy.
5. **Prediction:** Predict future rainfall and visualize the results.

2. Details of the City/Study area:

Attribute	Details
City Name	
State / Province	
Country	
Latitude	
Longitude	
Area (sq km)	
Climate Type	
Average Annual Rainfall (mm)	
Maximum Rainfall (mm/day)	

<Insert location map/google map of the city>

3. Basic Information of the Data

Description of the Dataset:

Source of the Data: https://power.larc.nasa.gov/data-access-viewer/

Example
NASA/POWER Source Native Resolution Daily Data
Dates (month/day/year): 01/01/1990 through 12/31/2024 in UTC
Location: latitude 9.1644 longitude 76.7423 elevation from MERRA-2: Average for 0.5 x 0.625 degree lat/lon region = 211.5 meters
The value for missing source data that cannot be computed or is outside of the sources availability range: -999

Parameters:

Attribute	Description
ALLSKY_SFC_SW_DWN	CERES SYN1deg All Sky Surface Shortwave Downward Irradiance ($MJ/m^2/day$)
ALLSKY_SFC_SW_DNI	CERES SYN1deg All Sky Surface Shortwave Downward Direct Normal Irradiance ($MJ/m^2/day$)
T2M	MERRA-2 Temperature at 2 Meters (°C)
T2M_MIN	MERRA-2 Temperature at 2 Meters Minimum (°C)
T2M_MAX	MERRA-2 Temperature at 2 Meters Maximum (°C)
QV2M	MERRA-2 Specific Humidity at 2 Meters (g/kg)
RH2M	MERRA-2 Relative Humidity at 2 Meters (%)
PRECTOTCORR	MERRA-2 Precipitation Corrected (mm/day)
WS2M	MERRA-2 Wind Speed at 2 Meters (m/s)
GWETTOP	MERRA-2 Surface Soil Wetness (1)

- **Target Variable:**
  - Daily Rainfall (mm): PRECTOTCORR

4. Splitting of data: YEAR WISE

Training & Testing period: 1995 to 2020. Split data year wise for a ratio of 80:20.

Example: Training: (80%) 1995,1998,1990,2000,1996,2002,etc

Testing: (20%): 1997,1991,1997…..

Unseen data/ prediction: 2021 to 2025

- **These are actual rainfall values.**
- **Do not use them for training or testing your model.**
- **Use the best model to predict rainfall for this period.**

- **Compare your predicted values with these actual values to evaluate your model's accuracy.**

5. Data Cleaning Report:

Handling Missing Values:

(Describe how missing values were handled, e.g., imputation, removal of rows/columns, etc.)

Outlier Detection and Removal:

(Explain how outliers were identified and handled, if applicable)

Data Normalization/Standardization:

(If any scaling was performed, mention the methods, e.g., Min-Max Scaling, Z-score Standardization)

Final Dataset:

(Provide a brief summary of the cleaned dataset, including the number of rows/columns after cleaning)

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6. Feature Ranking Report

Feature Importance:

(Explain how you identified which features are most relevant for predicting rainfall. Mention methods like correlation analysis, mutual information, or feature importance techniques.)

Top Features:

(List the most important features that significantly contribute to predicting rainfall)

7. Principal Component Analysis (PCA) Report

PCA Objective:

(Explain the purpose of performing PCA—reducing dimensionality, visualizing variance, etc.)

Results of PCA:

(Summarize the results, such as the number of principal components chosen and the explained variance ratio)

Visualizations:

(Include any graphs or plots like PCA biplots or variance explained by each component)

8. Development of ML models

- **Models Used:**
  - (Describe the machine learning models you have tried. For example, Decision Trees, Random Forest, Support Vector Machines, KNN, Neural Networks, etc.)
- **Reason for Choosing Each Model:**
  - (Justify why each model was chosen for the task)
- **Best Performing Model:**
  - (Which model gave the best results and why?)
- **Table Comparison:**
  - Provide a table comparing the performance of different models based on R^2, RMSE, and accuracy on both training and testing data.

Model	R^2 Score (Train)	R^2 Score (Test)	RMSE (Train)	RMSE (Test)	MAE (Train)	MAE (Test)	MSE (Train)	MSE (Test)
Model 1 (e.g., RF)								
Model 2 (e.g., SVM)								
Model 3 (e.g., NN)								
…								
….								
…								
…								
…								
…								

7. Selection of best ML model

- **Selected Model:**
  - (Describe the model that gave the best results based on the comparison, including why it was chosen)
- **Evaluation Metrics:**
  - Discuss the evaluation metrics used, such as R^2 score, RMSE, accuracy, etc.
- **Performance Indicators:**
  - Provide a table comparing the model's performance on the training and test sets.

Best ML Model	Metric	Training Set	Test Set
	R^2 Score		
	RMSE		
	MAE		
	MSE		

9. Prediction with the best model for the year 2025

- **Predicted Rainfall vs Actual Rainfall:**
  - (Provide a table with actual vs predicted values for the test set)
- **Prediction Performance:**
  - (Discuss how well the model performed in terms of prediction accuracy, e.g., the predicted rainfall vs. observed rainfall)

- **Graphs and Visualizations:**
  - Include relevant visualizations, such as:
    - **Time-series plots** of rainfall over time.
    - **Predicted vs. Actual** rainfall comparison plots.
    - **Model performance** graphs (e.g., learning curves, residual plots).
  - Provide a brief analysis of each graph.

Example

Date	Actual Rainfall (mm) (Unseen data)	Predicted Rainfall (mm) (by the best model)
01-01-2021		
02-01-2021		
03-01-2021		
…		
…..		
….		
31-12-2025		

10. Create a pickle file to predict rainfall using the best model:

- Save your best-trained model as a pickle file.
- This pickle file will allow the user to predict rainfall for a specific day.
- The user will enter values for 11 features corresponding to that day.
- The model should then output the predicted rainfall based on the entered values.
- This has to be demonstrated to the course coordinators

(Insert screenshot of the pickle interface or page here)

11. Conclusion

- **Discuss ML applications in prediction of rainfall**
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Annexures:

1. **Complete Python Code:**
   - Attach the full code used for data pre-processing, feature engineering, model training, evaluation, and predictions.
2. **Cleaned Dataset:**
   - Attach the cleaned dataset that was used in the analysis.

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